

CROSS-PLATFORM MOBILE APP DEVELOPMENT USING FLUTTER PROJECT SHOWCASE

Comprehensive project portfolio mapping creative concept, engineering pipeline, deliverables, team, and ecosystem impact.

1 PROJECT IDENTITY

PROJECT NAME AVO HealthCare Your Connected Health Ecosystem	JOB PROFILE Mobile App Dev Flutter Clean Arch & MVVM	SUPERVISOR Eng. Ibrahim Saad DEPI Program Lead & Mentor
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2 TEAM (Members & Roles)

 Yehia Mahmoud Team Lead & AI/Reminders	 Mamdouh Salah Admin, Doctor & Chat UI	 Abdallah Mahmoud Auth & Multi-Role Sync
 Ziad Magdy Appointment Cycle Dev	 Abdelrhman Saleh Profile & Health Records	 Rehab Hatem UI/UX, Pharmacy & Docs

3 CREATIVE IDEA (The A + V + O Concept)

Concept Centralized smart healthcare platform uniting Patients, Doctors, Pharmacies & Labs with an Admin governance layer.
Inspiration Overcoming Egypt's fragmented paper records, unverified providers, pharmacy dispatch delays, and diagnostic access barriers.
Unique Value (A + V + O) Automate e-Rx & booking, Visualize AI test OCR & analytics, Optimize care with Firestore & Hive.

4 PROBLEM & AUDIENCE (5 Stakeholders)

Challenge Fragmented health records, unverified practitioner accounts, delayed emergency access & manual pharmacy fulfillment.
Target Users & User Needs Patients (booking, e-Rx, Al triage), Doctors (EHR, e-Rx), Pharmacies (orders), Labs (reports) & Admin (licensing & logs).

FLUTTER MOBILE SUITE



Clean Architecture

MVVM & Cubit

Firestore Streams

\$ Material 3

Offline Hive

5 CREATIVE PROCESS (5-Phase Lifecycle)

1 Research 5-actorscope & market study (Vezeeta, Yodawy).	2 Ideation Use-cases, DFD Level 0/1 & Admin gatekeeping.	3 Design Figma design system, dual themes & prototypes.	4 Production Clean Arch, MVVM Cubit, Firestore & Hive cache.	5 Testing Unit tests, widget tests, Play & TestFlight.
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6 TOOLS & TECHNOLOGIES

MOBILE	BACKEND	STORAGE	AI & APIS	DESIGN/UML
Flutter 3.x (Dart)	Firebase Auth	Hive (Offline)	Gemini LLM	Figma System
Clean Arch	Cloud Firestore	SharedPreferences	AI Lab OCR	Draw.io (UML)
Cubit / BLoC	Realtime DB	Cloudinary	Bilingual Chat	Inter Typeface
ScreenUtil	FCM Alerts	Secure Storage	PayPal / Apple	Dual Theme

7 DELIVERABLES

UI Design System Figma atomic system, custom teal tokens (#00A991, #00876B), and role-tailored UI for 5 stakeholders.	5-Role Flutter App Working Flutter app with 5 role modules: Patient, Doctor, Pharmacy, Laboratory & Admin Dashboard.	AI Health Assistant AI-powered medical report scanner (OCR text extraction) & conversational bilingual health chatbot.	Rollout & Docs 36-page technical documentation, complete UML system diagrams, Google Play & TestFlight builds.
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8 Technical & Impact KPIs

< 2.0s Fast Load Instant 4G render	60 FPS Smooth UI Zero frame drops	99.5% Availability Hive offline sync	5 ROLES Unified Eco Live stream sync	100% Verified Syndicate gatekeep
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9 Security & Compliance Matrix

Firestore RBAC Granular role-based security rules preventing cross-actor data exposure.	TLS 256-bit End-to-end encrypted packet transmission for all live database streams.	Syndicate Gate Mandatory Admin audit of doctor/pharmacy syndicate licenses before activation.	Secure Storage Encrypted SharedPreferences & secure key storage for tokens & biometric sessions.
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10 Key App Innovations & Smart Services

Gemini OCR Camera digitizer converting paper lab tests into structured blood analytics.	Med Reminders Background dose alarm scheduler with persistent intake adherence logging.	Bilingual AI Seamless Arabic (RTL) & English (LTR) with instant localized triage chat.	Live Timers Synchronized consultation duration timer & real-time queue trackers.
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Product & Feature Highlights

